THE UNIVERSITY OF HONG KONG

BACHELOR OF ENGINEERING EXAMINATION

DEPARTMENT OF CIVIL ENGINEERING

CIVL1105: Environmental Engineering

May 13, 2016 TIME: 2:30 pm – 5:30 pm

I. Answer ALL questions. Section A carries 30 marks, and Section B carries 70 marks.

II. Atomic weights: H 1; C 12; N 14; O 16; F 19; Na 23; Mg 24.3; Al 27; P 31; S 32; Cl 35.5; K 39; Ca 40; Cr 52; Mn 55; Fe 55.8; Ni 58.7; Cu 63.5; Zn 65.4; I 127.

III. Properties of water:

Density = 1000 kg/m^3
Viscosity (15 °C) = 1.1404 mPa·s
Specific heat = 4.18 kJ/kg.°C
Heat of vaporization (15 °C) = 2465 kJ/kg
Heat of fusion = 333 kJ/kg

Use of Calculators:

Only approved calculators as announced by the Examinations Secretary can be used in this examination. It is candidates' responsibility to ensure that their calculator operates satisfactorily, and candidates must record the name and type of the calculator used on the front page of the examination script.

Section A. (10 multiple choice questions with 3 marks each)

1. A three-way catalytic converter controls the emission of (a) NO_x, CO_2, and HC; (b) NO_x, CO_2, and O_3; (c) NO_x, CO, and HC; (d) NO_x, CO, and O_3; or (e) none of the above.
2. Methane has an energy content of 35.8 kJ/L (@STP). Estimate the amount of heat released by burning 1 kg methane: (a) 2.2 kJ, (b) 2.2 MJ, (c) 35.8 kJ, (d) 35.8 MJ, (e) 50 kJ, or (f) 50 MJ.
3. The biomass content in an activated sludge process is best represented by (a) the total solids, (b) the volatile dissolved solids, (c) the dissolved solids, (d) the fixed solids, or (e) the volatile suspended solids.
4. Which of the following mechanisms is not involved in coagulation treatment? (a) Charge neutralization, (b) charge reversal, (c) bridging, (d) entrapment, or (e) none of the above.
5. Which biological treatment is an attached aerobic treatment? (a) Activated sludge, (b) aerated lagoon, (c) trickling filter, (d) sludge digester, (e) membrane bioreactor, or (f) none of the above.
6. Which of the following treatment processes is not intended for particle removal from flue gas? (a) Cyclone, (b) gravity settler, (c) scrubber, (d) baghouse, or (e) electrostatic precipitator.
7. Which of the following water quality parameters is not directly related to the organic strength of wastewater? (a) theoretical oxygen demand, (b) TOC, (c) COD, (d) NBOD, or (e) none of the above.
8. Which compound is not directly related to greenhouse effect? (a) CO_2, (b) water vapor, (c) CH_4, (d) O_3, (e) CFCs, or (f) none of the above.
9. Denitrification refers to (a) the removal of nitrogen compounds by chemical oxidation, (b) the biochemical conversion of ammonia to nitrate, (c) the biochemical conversion of nitrate to nitrogen gas, (d) the biochemical conversion of ammonia to nitrogen gas, (e) the physical removal of volatile ammonia by air stripping, or (f) the chemical precipitation of nitrogen compounds by forming insoluble solids.
10. Which of the following is the best estimate for the present concentration of CO_2 in ambient air? (a) 0.75 g/m^3, (b) 7.5 g/m^3, (c) 750 g/m^3, (d) 7,500 g/m^3, (e) 75,000 g/m^3, or (f) 750,000 g/m^3.

Section B (Three questions with unequal marks)

11. Briefly explain the following concepts:

    (a) Oxygen sag curve

    (6 marks)

    (b) Temperature inversion

    (6 marks)

12. A ground water (at 25°C) has reached equilibrium with limestone $CaCO_3(s)$. The equilibrium partial pressure of CO_2 is 0.001 atm, and the pH of the ground water is 8.

(a) Determine the molar concentration of HCO_3^-, CO_3^{2-}, and Ca^{2+}.

(12 marks)

(b) Determine the alkalinity, total hardness, and carbonate hardness. Express your results in **mg as $CaCO_3(s)$/L**.

(8 marks)

(c) Without going through any calculation, briefly discuss what will happen if the partial pressure of CO_2 increases.

(5 marks)

You may need to use the following thermodynamic data in your calculations:

- Henry's constant for CO_2 at 25°C: $K_H = 3.39 \times 10^{-2}$ M/atm
- Acidity constants of carbonic acid (H_2CO_3) at 25°C: $pK_1 = 6.35$ and $pK_2 = 10.33$
- Solubility of $CaCO_3(s)$: $pK_{so} = 8.3$

13. A coal-fired 20-MW power plant has an energy efficiency of 33%. The properties of coal used in the power plant are given below:
    - Energy content: 24 MJ/kg coal
    - Carbon content: 62%
    - Sulfur content: 1%
    - Ash content: 10% (with 75% of the ash emitted as particulate matter through flue gas)

(a) Determine the daily amount of coal consumption, carbon emission, SO_2 emission, and particulate matter emission for the plant.

(12 marks)

(b) Briefly discuss the environmental and health effects of SO_2. How to control SO_2 emission from the plant?

(8 marks)

(c) The plant uses cooling water drawn from a nearby river. It is important to ensure that the temperature increase of the cooling water is no more than 10 °C. What is the required flow rate of cooling water? You can assume 85% of the waste heat from the plant is removed by the cooling water and the remaining 15% released to the atmosphere through flue gas.

(8 marks)

(d) An alternative fuel for power generation is natural gas. Do you think natural gas is a cleaner fuel? Justify your answer with brief explanation.

(5 marks)

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